

1 · DEFINE & EXCLUDE MIMICS

Syncope = abrupt transient LOC, global cerebral hypoperfusion, rapid full recovery (6–8 s of ↓flow suffices). Subset of **TLOC** (short, abnl motor control, unresponsive, amnesia).

Mimic	Tell-tale
Seizure	Prolonged convulsions, post-ictal confusion, lat. tongue bite, incontinence (convulsive syncope = brief jerks, <u>no</u> post-ictal state)
TIA/CVA	Posterior circ. — LOC only <u>with</u> focal signs (vertigo, dysarthria); never isolated
Hypoglycemia	Impaired (not lost) consciousness; longer; no recovery w/o Rx
Other	Intoxication, psychogenic pseudosyncope, concussion

2 · THREE CLASSIFICATIONS

Cascade: **CPP = MAP – ICP**; MAP = CO × SVR; CO = HR × SV (SV ← preload, afterload, contractility). Every cause is a lesion in this loop.

Type	Defect / causes
Orthostatic (insufficient response)	↑SBP ≥20 or ↓DBP ≥10 on standing. Sympathetic (neurogenic: Parkinson/MSA, DM, amyloid) · SVR (drugs = #1 cause) · HR (BB/CCB) · preload (volume loss, diuretics, hemorrhage)
Reflex (inappropriate response)	Vasovagal (orthostatic/emotional) · situational (micturition, defecation, cough, post-exercise) · carotid sinus hypersens. Subtypes: cardioinhibitory (↓HR), vasodepressive (↓SVR), mixed
Cardiac (the one that kills)	↑Afterload (AS, HOCM) · ↓contractility (MI, VT) · ↓HR (AVB, SND, PM malfx) · ↑preload (PE, tamponade, pulm HTN, dissection)

PE myth: 17% in selected hospitalized cohort (Prandoni 2016), but <1% of all comers & <3% hospitalized (2018, >1.5M pts). Uncommon — don't CT every fainter.

3 · EVALUATION — HISTORY DOES THE WORK

~60% of TLOC sorted by H&P alone. **Who / Where-how / Prodrome:**

	OH	Reflex	Cardiac
Who	Older, comorbid, autonomic	Any age, healthy, recurrent <40	Structural HD/CAD, FHx SCD
Where	On/after standing, post-prandial	Standing, hot/crowded, trigger	Supine/rest (arrhythmia), exertion (structural)
Prodrome	Lightheaded w/ posture	Pallor, diaphoresis, nausea	None, or sudden palpitations

Everyone gets 4: History · exam (cardiac) · orthostatic vitals (within 3 min) · 12-lead ECG.

3b · TARGETED TESTING

- OH:** BMP/CBC (volume, occult bleed); autonomic eval if neurogenic (Valsalva w/o HR/BP rise)
- Reflex:** hx dx; tilt-table if unclear (cardioinhibitory predicts asystolic syncope); carotid massage in >40 w/ suggestive hx (**contra:** bruit, stenosis >70%, TIA/CVA <3 mo)
- Cardiac:** tele, TTE (if structural susp.), ±trop/BNP, stress if exertional + CAD risk, CTPA only if PE susp.

Monitor	Dur.	For
Holter	24–72h	Daily–weekly events
Event/loop	2–6 wk	Mod. freq + prodrome
Patch	~2 wk	Mod. freq, comfortable
ILR	Years	Infrequent/unexplained (highest yield)

AVOID in simple syncope (nl neuro, no trauma, no szr features): head CT/ MRI, EEG, carotid US — yield ~0.2–1.5%; routine TTE/CTPA/broad labs.

4 · DISPOSITION — HIGH-RISK = ADMIT

	Low → home	High → admit
PMH	Recurrent low-risk; no structural HD	Structural HD/CAD; FHx SCD
Event	Typical reflex prodrome; clear trigger	Exertional/supine; pre-event palpitations; short/absent prodrome (<10s) w/ structural dz
Exam	Normal	SBP<90; awake HR<40; GIB signs; new murmur
ECG	Normal	Ischemia; Mobitz II/3° AVB; pauses>3s; VT; pre-excit (WPW); QTc<340 or >460; Brugada; ARVC/epsilon; BBB/Q/LVH

Neither clearly high nor low → ED/obs.

CSRS — 30-DAY SERIOUS ADVERSE EVENT

Category (score)	30d SAE	Dispo
Very low (–3 to –2)	0.3–0.4%	D/C, RTC
Low (–1 to 0)	0.7%	D/C, RTC
Medium (1 to 3)	~8%	Brief obs / shared decision
High (4 to 5)	~20%	Tele / admit
Very high (≥6)	~51%	Tele / admit

Points: vasovagal predispo –1 · hx heart dz +1 · ED SBP <90 or >180 +2 · trop >99%ile +2 · abnl QRS axis (<–30 or >100) +1 · QRS >130 ms +1 · QTc >480 ms +2 · ED dx vasovagal –2 · ED dx cardiac +2

5 · MANAGEMENT

- OH:** fluids, stop/adjust culprit meds, slow position change, salt, compression, counter-pressure maneuvers; midodrine/fludrocortisone if autonomic
- Reflex:** reassurance, trigger avoidance, counter-pressure if prodrome; ±midodrine/fludro/salt; pacemaker select (>40 + documented cardioinhibitory + frequent/unpredictable)
- Cardiac:** consult cardiology/EP — etiology-specific (EF ≤35% → ICD discussion)

2026: 2021 ESC pacing guideline + BIOSync RCT (77% RRR @2y) strengthened DDD-CLS pacing for >40 w/ severe recurrent cardioinhibitory reflex syncope + documented asystole; cardioneuroablation emerging (device-free).

ATTENDING DOTPHRASE

#Syncope: transient LOC. witnessed/unwitnessed, w/wo prodrome, exertional/at-rest/postural, w/wo injury. likely 2/2 * [reflex/vasovagal, orthostatic, cardiac (arrhythmic/structural), neuro-mimic, med/vol]. r/o serious cause [arrhythmia, ACS, AS/HOCM, PE, GIB, AAA, SAH]; excl mimics [szr, hypoglycemia, intox, TIA, pseudosyncope]. CSRS = * [vlow ≤–2 / low –1–0 / med 1–3 / high 4–5 / vhigh ≥6]. HDS/HDUS. - core w/u: H&P [prodrome, trigger, EXERTIONAL, palpitations pre-event, FHx SCD/<50yo, structural HD, QT/AVN/diuretic/antiHTN meds, lat tongue bite/post-ictal] + 12-lead ECG [ischemia, AVB, BBB, QTc, Brugada, WPW, epsilon/ARVC, LVH] + orthostatics [SBP ≥20 / DBP ≥10 w/in 3min] - CSRS pts: vasovagal predispo –1 / hx heart dz (CAD/AF/CHF/valv) +1 / ED SBP <90 or >180 +2 / trop >99%ile +2 / abnl QRS axis +1 / QRS >130ms +1 / QTc >480ms +2 / ED dx vasovagal –2 / cardiac +2 → 30d SAE: vlow 0.4% / low 0.7% / med ~8% / high ~20% / vhigh ~51% (low → d/c w/ RTC; high/vhigh → tele/admit) - targeted only: hs-trop + ECG if cardiac susp;